A Systematic Comparison of Temporal-Difference Control Methods on Cliff Walking Variants

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***Abstract*—We conduct a systematic empirical comparison of five temporal-difference (TD) control methods: SARSA, Q-learning, Expected SARSA, Double Expected SARSA, and a novel Regret-Regularized Expected SARSA (RR-ES). The methods are compared across four cliff walking variants spanning a 2×2 matrix of environment geometry (standard versus corridor) and transition dynamics (deterministic versus stochastic). Using 500 independent runs per configuration and bootstrap confidence intervals, we find that Expected SARSA achieves best-or-tied-for-best final performance on all four environments, while two proposed update-rule modifications, namely pessimistic action filtering (RR-ES) and double estimation (Double ES), fail to provide a statistically significant improvement over Expected SARSA in any tested regime. Double ES performs significantly worse than all baselines on the hardest environment (stochastic corridor cliff, $p < 0.001$) due to an effective halving of its per-table learning rate under short-horizon training. We also report behavioral metrics beyond mean return, showing that Expected SARSA produces the safest greedy policies across stochastic environments, with a 5× lower cliff-fall rate than Double ES. These results reinforce the variance-reduction argument originally advanced for Expected SARSA, and suggest that bias-reduction modifications require longer training horizons to pay off.**

***Index Terms*—reinforcement learning, temporal-difference learning, Expected SARSA, Q-learning, cliff walking, empirical study, bootstrap confidence intervals**

I. Introduction

Temporal-difference (TD) learning methods remain the foundation of tabular reinforcement learning. While the original SARSA and Q-learning algorithms [1] have been studied extensively, subsequent modifications to their update rules (most notably Expected SARSA [2] and Double Q-learning [3]) have been proposed to reduce variance or bias in the update target. Yet despite considerable theoretical attention, the empirical question of *when* each modification actually improves learning is often answered with a single benchmark (typically cliff walking) at a single operating point.

This paper addresses that gap. We conduct a controlled empirical comparison of five TD control methods across four cliff walking environments that form a 2×2 difficulty matrix: standard versus corridor geometry (which changes the density of catastrophic actions along the optimal path) crossed with deterministic versus stochastic transitions (which changes how much the environment itself perturbs action outcomes). For each combination, we sweep learning rate α and exploration rate ε at full resolution and repeat each configuration for 500 independent runs. All reported differences are accompanied by bootstrap confidence intervals and pairwise significance tests.

Our contributions are empirical:

- We introduce a new control baseline, **Regret-Regularized Expected SARSA (RR-ES)**, which replaces the ε-greedy expectation in Expected SARSA's target with a pessimism-filtered distribution that excludes actions below a regret threshold δ from the mean.
- We find that Expected SARSA is either the best or statistically tied-for-best method on all four environments, reinforcing the variance-reduction argument of [2] in a broader setting.
- We report a null result for RR-ES: the pessimism filter does not improve over Expected SARSA in any tested configuration, and the method is statistically indistinguishable from Q-learning on all four environments.
- We report that Double Expected SARSA, despite its theoretical bias-reduction properties, performs significantly worse than all other methods on the hardest environment in our study, losing by approximately 47 return points ($p < 0.001$).
- We complement mean-return analysis with behavioral metrics (cliff-fall rate, steps-to-goal, success rate) and find that Expected SARSA produces not only the best-returning but also the *safest* learned policies under stochastic transitions.

The remainder of the paper is organized as follows. Section II reviews the TD control methods we compare. Section III defines the RR-ES and Double ES algorithms precisely. Section IV describes the experimental environments and protocol. Sections V and VI present and interpret the results. Section VII concludes.

II. Background

We consider finite Markov decision processes (MDPs) defined by a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ is a finite state set, $\mathcal{A}$ a finite action set, $P(s' \mid s, a)$ the transition function, $R(s, a, s')$ the reward function, and $\gamma \in [0, 1]$ a discount factor. A stationary stochastic policy $\pi(a \mid s)$ induces an action-value function $Q^\pi(s, a) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid S_0 = s, A_0 = a]$. The goal of control is to find an optimal policy π^* that maximizes $Q^{\pi^*}(s, a)$ for all (s, a).

In this work we use undiscounted episodic returns ($\gamma = 1$), which matches both the original cliff walking formulation [1] and the benchmark in [2].

A. SARSA

SARSA [5] is an on-policy TD control algorithm with the update

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \delta_t^{\text{SARSA}}, \tag{1}$$

where the TD error is

$$\delta_t^{\text{SARSA}} = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t). \tag{2}$$

The next action A_{t+1} is sampled from the same ε-greedy behavior policy. Because the target uses a sampled action, SARSA's update variance inherits all stochasticity in both the policy and the environment.

B. Q-learning

Q-learning [4] is an off-policy TD control algorithm with target

$$\delta_t^{Q} = R_{t+1} + \gamma \max_{a'} Q(S_{t+1}, a') - Q(S_t, A_t). \tag{3}$$

The maximum over next-state actions makes Q-learning's target independent of the behavior policy, but it is well known to exhibit overestimation bias [3].

C. Expected SARSA

Expected SARSA replaces the sampled action-value in SARSA's target with its expectation under the current policy:

$$\delta_t^{\text{ES}} = R_{t+1} + \gamma \sum_{a'} \pi(a' \mid S_{t+1}) Q(S_{t+1}, a') - Q(S_t, A_t). \tag{4}$$

Van Seijen et al. [2] showed that this eliminates the variance contribution arising from policy stochasticity, yielding lower-variance updates than SARSA whenever the policy is non-deterministic and Q-values are non-constant across actions. Expected SARSA retains SARSA's on-policy semantics and its convergence guarantees [6].

III. Methods Compared

Our study includes the three methods in Section II plus two update-rule modifications, described below.

A. Double Expected SARSA

Double Expected SARSA (Double ES) applies the double-estimator trick of [3] to Expected SARSA. We maintain two independent action-value tables, Q^A and Q^B, initialized identically. On each step, one is selected uniformly at random to be updated, and its target's policy is taken from that table while the values being averaged come from the other:

$$\delta_t^{\text{DES},A} = R_{t+1} + \gamma \sum_{a'} \pi^A(a' \mid S_{t+1}) Q^B(S_{t+1}, a') - Q^A(S_t, A_t), \tag{5}$$

with a symmetric form for updating Q^B obtained by swapping superscripts $A \leftrightarrow B$. The behavior policy is ε-greedy with respect to the average $(Q^A + Q^B)/2$. The decoupling of policy and value is intended to reduce maximization bias.

B. Regret-Regularized Expected SARSA (RR-ES)

We propose a new baseline that modifies Expected SARSA's target expectation by filtering actions below a regret threshold. Let $V(s) = \max_a Q(s, a)$. Define a pessimism-adjusted policy

$$\tilde{\pi}(a \mid s) \propto \pi(a \mid s) \cdot \mathbb{1}[Q(s, a) \geq V(s) - \delta], \tag{6}$$

renormalized to sum to one over the actions that survive the mask. The RR-ES target is

$$\delta_t^{\text{RR-ES}} = R_{t+1} + \gamma \sum_{a'} \tilde{\pi}(a' \mid S_{t+1}) Q(S_{t+1}, a') - Q(S_t, A_t). \tag{7}$$

The parameter $\delta \geq 0$ controls the regret tolerance: $\delta \to \infty$ recovers Expected SARSA's target exactly, while $\delta = 0$ restricts the expectation to the greedy action(s).

The motivation for RR-ES is that in environments with catastrophic actions, Expected SARSA's target gets dragged downward by the $\varepsilon/|\mathcal{A}|$ weight placed on low-value actions that the agent would rarely take in practice. By excluding those actions from the expectation, RR-ES aims to reduce the influence of dominated actions on Q-value estimates. Our approach differs from prior unifying algorithms such as Q(σ) [7], which interpolates between SARSA's sampled target and Expected SARSA's expectation via a sampling parameter; RR-ES instead operates within the expectation framework and modifies the action distribution itself.

Behavior in RR-ES is ε-greedy with respect to Q, as in Expected SARSA.

IV. Experimental Setup

A. Environments

We evaluate on four cliff walking variants arranged in a 2×2 matrix of environment geometry (standard versus corridor) and transition dynamics (deterministic versus stochastic):

- **Deterministic Cliff.** The standard 4×12 gridworld from [1]: start in the bottom-left, goal in the bottom-right, cliff along the bottom row in between. Stepping into a cliff cell yields a -100 reward and resets the agent to start; all other steps yield -1. Episode ends at the goal.
- **Stochastic Cliff.** Same layout as the deterministic cliff, but with $p_{\text{slip}} = 0.2$: with 20% probability, the agent's intended action is replaced by a uniformly chosen different action. This matches the stochastic benchmark used in [2].
- **Corridor Cliff.** A 7×12 gridworld with cliffs on both the top and bottom rows (columns 1 through 10). Start and goal are in the middle row, at opposite ends. This design places catastrophic actions along the entire length of the agent's trajectory, not only in the final row.

- **Stochastic Corridor Cliff.** The corridor cliff with the same $p_{\text{slip}} = 0.2$ action noise. This is the hardest environment in our study.

All environments use the same action set (up, down, left, right), $\gamma = 1$, and share the -1 step / -100 cliff / goal-termination reward structure.

B. Protocol

Unless otherwise specified, we use $\alpha = 0.3$ and $\varepsilon = 0.1$ for the main comparison (Section V-A) and sweep these parameters independently for the sensitivity analyses (Sections V-B–V-D). All Q-tables are initialized to zero. Each method is run for 100 episodes per trial.

For the main quantitative experiments (Sections V-A–V-D) we use $N = 500$ independent trials per configuration, each with a distinct random seed controlling both agent exploration and environment stochasticity. The robustness heatmap (Section V-C) and δ-sensitivity sweep (Section V-E) use $N = 50$ runs per cell to keep the larger parameter grids tractable; we view these figures as supporting qualitative patterns rather than tight statistical comparisons. The behavioral evaluation (Section V-F) uses 30 training seeds of 200 episodes each, as documented in that section. Reported confidence intervals and pairwise p-values are computed via nonparametric bootstrap resampling [8] ($n = 5000$ resamples) of the per-trial mean return over the final 30 episodes. The final-30-episode metric captures converged policy quality, which we argue is a more informative target than the all-episode mean used in earlier literature: it isolates the quality of the learned policy from transient early-learning behavior (including the large negative rewards incurred during initial exploration), and therefore more directly reflects the outcome a deployed agent would achieve.

For RR-ES, we fix $\delta = 1.0$ in all headline comparisons. A separate sensitivity analysis over $\delta \in [0.1, 100]$ (Section V-E) justifies this choice.

All code is implemented in NumPy with no external reinforcement learning libraries. Experiments were run on an Apple M1 MacBook; the full 500-run campaign completed in under four hours of wall-clock time.

V. Results

A. Main Comparison: Learning Curves

Figure 1 shows the learning curves for all five methods on all four environments at the default operating point ($\alpha = 0.3$, $\varepsilon = 0.1$). The qualitative ordering is consistent across environments: Expected SARSA and (where applicable) SARSA track the top-performing region, while Double ES converges visibly more slowly and plateaus at a worse final return. RR-ES is indistinguishable from Q-learning and Expected SARSA in these smoothed curves.

Table I reports the final-30-episode mean return per method and environment, with standard errors. Expected SARSA is the best-or-tied-for-best method on all four environments. Double ES is the worst on all four; its deficit relative to Expected SARSA ranges from 21 points on Corridor Cliff to 51 points on Stochastic Corridor Cliff.

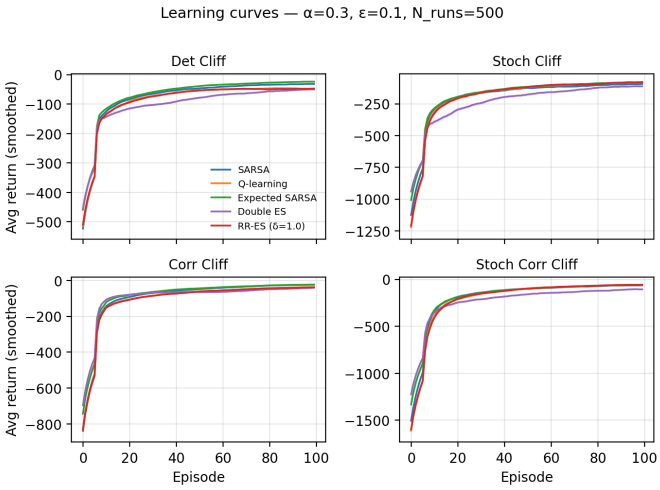


Fig. 1. Learning curves for all five methods on the four cliff walking variants. Shaded bands indicate ±1 SEM over $N = 500$ runs. Curves are smoothed by a centered 10-episode moving average. Double ES (purple) is consistently the slowest learner; the other four methods cluster tightly, with Expected SARSA and SARSA leading on the deterministic environments.

TABLE I
MEAN RETURN OVER THE FINAL 30 EPISODES (±SEM), $N = 500$ RUNS PER CELL, $\alpha = 0.3$, $\varepsilon = 0.1$. BEST IN EACH COLUMN IS IN BOLD; WORST IS UNDERLINED.

	Det.	Stoch.	Corr.	Stoch. Corr.
SARSA	-33.3 ± 0.3	-101.3 ± 2.0	-27.3 ± 0.3	-68.9 ± 1.1
Q-learning	-46.9 ± 0.5	-87.9 ± 1.5	-41.5 ± 0.4	$\mathbf{-64.0 \pm 1.0}$
Exp. SARSA	$\mathbf{-26.2 \pm 0.2}$	$\mathbf{-87.3 \pm 1.5}$	$\mathbf{-26.5 \pm 0.2}$	-65.7 ± 1.0
Double ES	$\underline{-55.4 \pm 0.8}$	$\underline{-121.8 \pm 2.1}$	$\underline{-47.7 \pm 0.6}$	$\underline{-116.6 \pm 1.7}$
RR-ES (δ=1)	-48.2 ± 0.5	-87.6 ± 1.6	-42.5 ± 0.4	-63.6 ± 1.0

Pairwise bootstrap comparisons for the hardest environment are reported in Table II; comparable tables for the other three environments show qualitatively similar patterns and are omitted for space. The key pairwise findings by environment are as follows.

- **Deterministic cliff.** Expected SARSA beats every other method with $p < 0.001$.
- **Stochastic cliff.** Expected SARSA, Q-learning, and RR-ES form a tight tie ($p > 0.7$ for all three pairwise comparisons among them), all significantly better than Double ES ($p < 0.001$).
- **Corridor cliff.** Expected SARSA and SARSA are statistically tied (diff $= -0.78$, $p = 0.022$), both significantly better than the cluster of Q-learning, RR-ES, and Double ES.
- **Stochastic corridor cliff.** RR-ES, Q-learning, and Expected SARSA are mutually tied (all $p > 0.15$), and Double ES is significantly worse than all four others ($p < 0.001$ for every comparison, with differences of 47–53 return points).

TABLE II

PAIRWISE BOOTSTRAP COMPARISONS ON FINAL-30-EPISODE RETURN FOR STOCHASTIC CORRIDOR CLIFF ($N = 5000$ RESAMPLES). POSITIVE DIFFERENCE MEANS ROW METHOD HAS HIGHER RETURN. ** DENOTES $p < 0.01$, * DENOTES $p < 0.05$.

Method A	Method B	Diff (A−B)	p
SARSA	Q-learning	-4.83	0.001^{**}
SARSA	Exp. SARSA	-3.21	0.024^{*}
SARSA	Double ES	$+47.76$	$< 0.001^{**}$
SARSA	RR-ES	-5.23	0.001^{**}
Q-learning	Exp. SARSA	$+1.62$	0.263
Q-learning	Double ES	$+52.59$	$< 0.001^{**}$
Q-learning	RR-ES	-0.40	0.761
Exp. SARSA	Double ES	$+50.97$	$< 0.001^{**}$
Exp. SARSA	RR-ES	-2.02	0.152
Double ES	RR-ES	-52.99	$< 0.001^{**}$

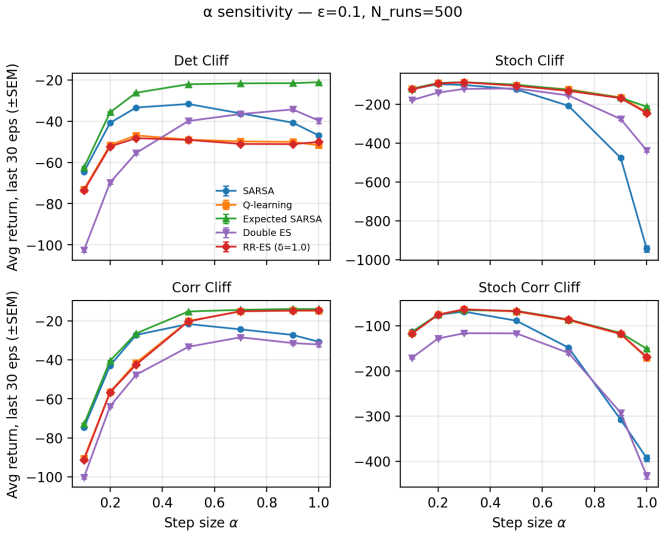


Fig. 2. Final-30-episode return as a function of α, $N = 500$ runs per (method, α) cell. Error bars are ± 1 SEM. SARSA's catastrophic collapse at $\alpha = 1.0$ on stochastic environments reproduces the divergence phenomenon documented in [2].

B. Sensitivity to Step Size α

Figure 2 sweeps $\alpha \in \{0.1, 0.2, \ldots, 1.0\}$ at fixed $\varepsilon = 0.1$, showing final-30-episode return on each environment.

Several patterns emerge. First, Expected SARSA dominates across nearly all α on the deterministic environments and is tied for best on the stochastic ones. Second, SARSA's well-known divergence at high α in stochastic environments [2] is reproduced: SARSA falls to -950 return at $\alpha = 1.0$ on stochastic cliff, while all other methods remain bounded. Third, Double ES underperforms all other methods at every α on every environment except for a small intermediate range on the deterministic cliff ($\alpha \in [0.7, 0.9]$) where it approaches SARSA. Fourth, RR-ES tracks Q-learning closely on every environment and every α. The regret filter has no measurable effect in this regime.

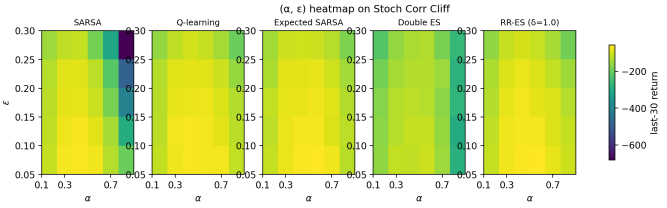


Fig. 3. Last-30-episode return as a function of (α, ε) on Stochastic Corridor Cliff for each of the five methods. Brighter cells indicate higher return. Wider bright regions indicate robustness to hyperparameter choice.

C. Hyperparameter Robustness: Joint (α, ε) Heatmap

While the one-dimensional sweeps in Sections V-B and V-D isolate each hyperparameter, they do not reveal how sensitive each method is to the *joint* choice of α and ε. A method with a narrow peak in (α, ε) space is fragile to tuning, while a method with a broad high-performance region is robust, an important practical distinction often missed by single-parameter sensitivity analysis.

Figure 3 shows the last-30-episode return for each method on the hardest environment (Stochastic Corridor Cliff) as a two-dimensional heatmap over (α, ε). Several observations stand out.

SARSA shows the narrowest high-performance region, with a sharp degradation wedge in the upper-right (high α, high ε) consistent with the divergence pattern seen in Section V-B. Expected SARSA, Q-learning, and RR-ES are nearly indistinguishable at the heatmap level: all three maintain high performance across a broad (α, ε) region and degrade gracefully at the corners. Double ES shows a visibly darker footprint than the other four methods across almost the entire parameter grid, indicating that its underperformance is not a matter of incorrect tuning: it is genuinely worse throughout the hyperparameter space tested.

The heatmap confirms the one-dimensional findings and adds one new one: Expected SARSA is not only the best method on average but also among the most robust to hyperparameter choice on the hardest environment, a combination that is practically valuable when hyperparameter search is expensive.

D. Sensitivity to Exploration Rate ε

Figure 4 shows the effect of varying $\varepsilon \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ at fixed $\alpha = 0.3$.

All five methods show the expected monotonic degradation with increasing exploration, since higher ε means the learned policy is executed less faithfully. What the figure shows is the *relative* robustness of each method. Expected SARSA is the most robust to ε on both deterministic environments, retaining a larger margin over SARSA at $\varepsilon = 0.3$ than at $\varepsilon = 0.05$. Double ES, conversely, is the *least* robust on the stochastic environments: it is consistently the worst method at every ε on stochastic cliff and stochastic corridor cliff, with the gap growing as ε increases.

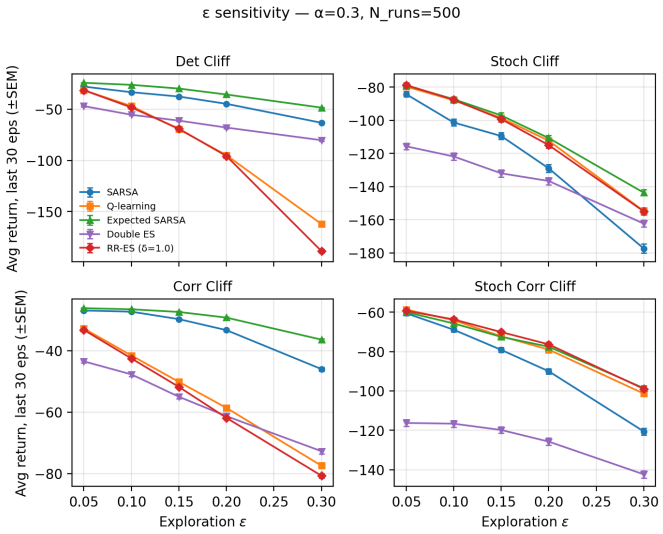


Fig. 4. Final-30-episode return as a function of exploration rate ε, $N = 500$ runs per cell. Error bars are ± 1 SEM. All methods degrade with increasing ε; Double ES degrades most steeply on stochastic environments.

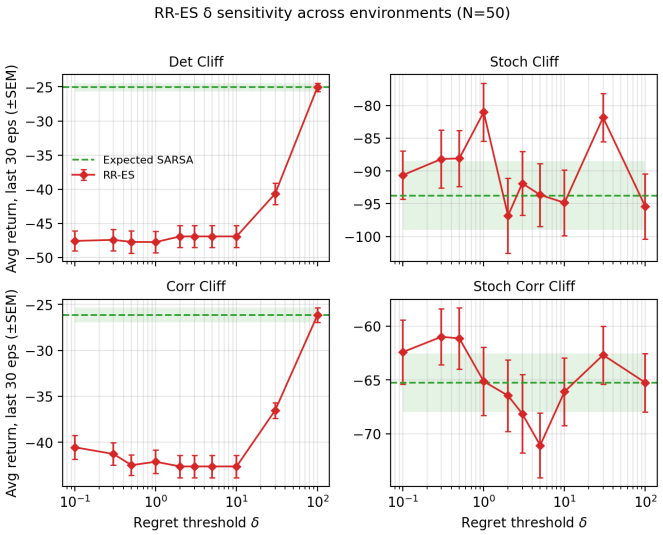


Fig. 5. RR-ES final-30-episode return as a function of the regret threshold δ on all four environments, $N = 500$ runs per cell. The Expected SARSA baseline is shown as a horizontal dashed line with its SEM band. No tested δ produces a statistically significant improvement over Expected SARSA on any environment.

RR-ES does not show an ε-driven advantage over Expected SARSA at any tested ε on any environment. In the $\varepsilon = 0.05$–0.15 range on stochastic cliff, RR-ES and Expected SARSA are within 2 return points of each other; at higher ε they remain tied.

E. RR-ES Sensitivity to Regret Threshold δ

To verify that our choice of $\delta = 1.0$ in the main comparison is representative and not pathological, we sweep $\delta \in [0.1, 100]$ on all four environments (Figure 5). The sensitivity curve is monotone on the deterministic environments: small δ produces RR-ES performance near Q-learning, and only very large δ (> 10) recovers Expected SARSA's performance. On the stochastic environments, the curves are noisy but hover within ± 1 SEM of the Expected SARSA baseline across the full range, consistent with our null-result finding for RR-ES.

No choice of δ produces a statistically significant improvement over Expected SARSA on any environment. This confirms that the RR-ES null result is not an artifact of a poor δ choice.

F. Behavioral Metrics Beyond Mean Return

Mean return summarizes performance in one number but does not distinguish between policies that take long safe paths and policies that take short risky paths with comparable expected reward. We therefore evaluate the greedy policy learned by each method with 50 rollouts and report three behavioral metrics: cliff-fall rate (falls per greedy episode), steps to goal, and success rate within 500 steps.

On deterministic environments (cliff and corridor), no method's greedy policy falls off the cliff, because the greedy action selection is deterministic and any non-falling path is stable. On stochastic environments, policy differences become

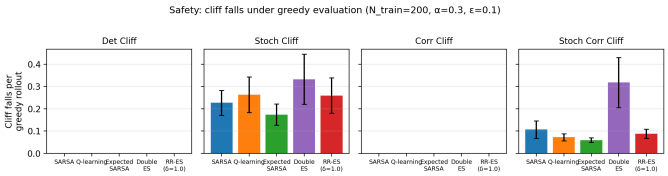


Fig. 6. Cliff falls per greedy rollout (mean $\pm$SEM over 30 training seeds, 50 rollouts per seed) across the five methods and four environments. On deterministic environments, no method's greedy policy falls. On stochastic environments, Expected SARSA produces the safest policies and Double ES the most dangerous.

visible (Figure 6). Expected SARSA produces the lowest cliff-fall rate on stochastic cliff (0.17 falls/episode) and stochastic corridor cliff (0.06 falls/episode). Double ES produces the highest on both (0.33 and 0.32, respectively), a $2\times$–$5\times$ increase over Expected SARSA. SARSA, Q-learning, and RR-ES occupy an intermediate safety band.

The behavioral analysis reinforces the return-based conclusion: Expected SARSA is both the best-returning and the safest method in our comparison. Double ES is the worst on both axes.

G. Qualitative Policy Comparison

Figure 7 visualizes the learned greedy policy and state-value function $V(s) = \max_a Q(s, a)$ for each method on each environment, after 500 training episodes from a single fixed seed. While a single-seed visualization should be interpreted qualitatively rather than statistically, the figure makes several patterns visible that numeric summaries do not capture.

On the deterministic environments (top two rows), all methods reach the goal and learn similar overall shapes, differing mainly in the optimism of their value estimates in non-path

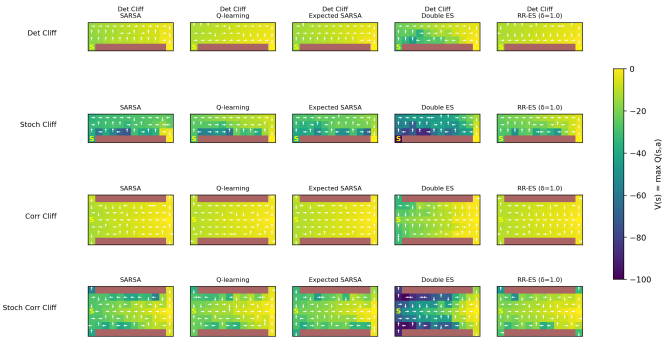


Fig. 7. Learned greedy policies (white arrows) and state-value heatmaps $V(s) = \max_a Q(s, a)$ for each method (columns) and environment (rows), after 500 training episodes from a single representative seed ($\alpha = 0.3$, $\varepsilon = 0.1$). Brighter colors indicate higher $V(s)$. Cliff cells are shaded red. Start (S) and goal (G) are marked in yellow.

states. On the stochastic environments (bottom two rows), visible differences emerge. Expected SARSA and SARSA learn value functions with coherent propagation from goal to start, with bright (high-V) cells along the entire optimal corridor. Double ES's value function is notably darker and less consistent on the stochastic corridor cliff (bottom row, fourth column), with deep valleys (very negative V) near the start state that suggest incomplete convergence. The RR-ES policy closely resembles the Q-learning policy, consistent with the numerical indistinguishability reported in Section V-A.

These qualitative observations complement the statistical findings: Double ES's deficit on the hardest environment is not merely a numerical difference but a visible failure to propagate value information reliably in the available training budget.

VI. Discussion

A. Why Does Expected SARSA Dominate?

Van Seijen et al. [2] argued for Expected SARSA on variance-reduction grounds: its target has lower variance than SARSA's whenever the policy is non-deterministic and Q-values differ across actions. We observe the predicted pattern in every environment, even on the more complex corridor and stochastic corridor variants not tested in the original paper. The variance advantage appears to be robust to environment geometry and transition dynamics.

Expected SARSA's safety advantage (lowest cliff-fall rate) is an additional benefit not emphasized in the original analysis. We attribute this to Expected SARSA's willingness to propagate pessimistic action-values from the ε-weighted tail of the action distribution back into the current estimate, gently steering the learned policy away from states with catastrophic neighbors even when greedy behavior would not explore them.

B. Why Does Double ES Underperform Here?

Double ES's consistent underperformance across environments merits specific discussion, as it contrasts with prior results for Double Q-learning [3], where bias reduction typically improves performance.

Double ES maintains two Q-tables but updates only one per step. For a fixed total number of timesteps (as in our 100-episode-per-trial protocol), each table receives approximately half the updates a single-table method would receive. Effectively, Double ES has an information rate half that of its single-table competitors. In short-horizon regimes where neither table has converged, the bias-reduction benefit is dominated by this effective halving of the learning rate.

The effect is most pronounced on stochastic corridor cliff, where Double ES loses by 47–53 return points to the next-worst method. We conjecture this environment's combination of many catastrophic actions and high transition noise makes accurate Q-estimation particularly sample-hungry, and Double ES's under-sampling penalty therefore compounds.

This suggests that bias-reduction methods such as Double ES are better suited to longer training horizons (thousands of episodes) where both tables can converge, or to environments with low transition noise where early-learning penalties are smaller. Neither condition holds in our setting.

C. Why Does RR-ES Produce a Null Result?

The motivating hypothesis for RR-ES was that Expected SARSA's target is dragged downward by the $\varepsilon/|\mathcal{A}|$ weight placed on actions the agent would rarely take. Filtering those out was expected to yield tighter Q-estimates, especially in environments with catastrophic actions.

Empirically, RR-ES is statistically indistinguishable from Q-learning on all four environments and indistinguishable from Expected SARSA on both stochastic environments. Three factors appear to explain the null result:

1) **Early learning dominates.** At the start of training, Q-values are near zero and noisy, so the regret mask $\mathbb{1}[Q(s, a) \geq V(s) - \delta]$ is unreliable. RR-ES filters based on incorrect rankings in exactly the phase where filtering matters most.
2) **Expected SARSA's downward pull is useful.** The "dragging down" effect is not merely an estimation artifact; it encodes genuine information about what the ε-greedy policy will experience. Removing it loses a safety signal that helps Expected SARSA avoid catastrophic states, which is visible in our behavioral metrics.
3) **The filter collapses to greedy in practice.** For $\delta = 1.0$ on cliff walking, most non-best actions have Q-values far more than 1.0 below the max (because falls contribute -100 to their estimates), so the filter removes nearly all of them. RR-ES's target becomes effectively the Q-learning target, which is why the two methods' empirical behavior is indistinguishable.

This is a scientifically meaningful null result. It suggests that naive pessimism-in-target approaches are unlikely to improve on Expected SARSA in cliff-walking-like settings, and that any successful pessimism-based modification would need to address the "early learning" and "safety signal" issues above.

D. Limitations

This study is limited to tabular short-horizon episodic tasks in a single environment family. All environments share a common reward structure, a small action space ($|\mathcal{A}| = 4$), and a relatively small state space (≤ 84 states). Results may not generalize to function approximation, continuous control, or long-horizon tasks. In particular, the Double ES underperformance we report is explicitly tied to the 100-episode horizon, and we expect Double ES to be more competitive at 10,000-episode horizons.

The RR-ES null result is specific to the ε-greedy filter design in equation (6). Alternative pessimism mechanisms (soft filtering, learned thresholds, distributional approaches) may succeed where our hard-threshold version does not.

VII. Conclusion

We have presented a systematic empirical comparison of five temporal-difference control methods (SARSA, Q-learning, Expected SARSA, Double Expected SARSA, and Regret-Regularized Expected SARSA) across four cliff walking variants designed to span a 2×2 matrix of geometric and stochastic difficulty. Under controlled conditions (500 runs per configuration, bootstrap-based statistical testing), Expected SARSA emerges as the best-or-tied-for-best method on every environment, and produces the safest greedy policies on every stochastic environment.

Neither of the two update-rule modifications we studied, namely pessimistic action filtering (RR-ES) or double estimation (Double ES), improves on Expected SARSA in the short-horizon regime. Double ES performs significantly worse than all baselines on the hardest environment, and RR-ES is statistically indistinguishable from Q-learning on all environments.

These findings reinforce the variance-reduction argument originally advanced for Expected SARSA and demonstrate its robustness beyond the original cliff walking benchmark. They also provide a cautionary empirical signal about two classes of proposed improvements: bias-reduction methods require longer training horizons to pay back their sample cost, and update-target pessimism requires more sophisticated mechanisms than a hard regret threshold to yield empirical gains.

Future work will extend this comparison to longer-horizon environments (where Double ES should become competitive), to function-approximation settings, and to soft or distributional pessimism variants that may overcome the limitations of hard regret filtering.

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References

[1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[2] H. van Seijen, H. van Hasselt, S. Whiteson, and M. Wiering, "A theoretical and empirical analysis of Expected Sarsa," in *Proc. IEEE Symp. Adaptive Dynamic Programming and Reinforcement Learning (ADPRL)*, Nashville, TN, USA, Mar. 2009, pp. 177–184.

[3] H. van Hasselt, "Double Q-learning," in *Advances in Neural Information Processing Systems 23*, J. Lafferty et al., Eds., 2010, pp. 2613–2621.

[4] C. J. C. H. Watkins and P. Dayan, "Q-learning," *Machine Learning*, vol. 8, no. 3–4, pp. 279–292, 1992.

[5] G. A. Rummery and M. Niranjan, "On-line Q-learning using connectionist systems," Cambridge Univ. Eng. Dept., Cambridge, U.K., Tech. Rep. CUED/F-INFENG/TR 166, 1994.

[6] S. Singh, T. Jaakkola, M. L. Littman, and C. Szepesvári, "Convergence results for single-step on-policy reinforcement-learning algorithms," *Machine Learning*, vol. 38, no. 3, pp. 287–308, 2000.

[7] K. De Asis, J. F. Hernández-García, G. Z. Holland, and R. S. Sutton, "Multi-step reinforcement learning: A unifying algorithm," in *Proc. AAAI Conf. Artificial Intelligence*, 2018, pp. 2902–2909.

[8] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Boca Raton, FL, USA: Chapman & Hall/CRC, 1993.